

The Devil's Advocate

Agentic SLM debate sparring partner

Overview and goal

This project will be an agent-based SLM application designed to act as a debate sparring partner. The core functionality is the following: a user inputs a thesis, opinion or decision and the agent automatically takes the opposing viewpoint to start a debate.

Using an agent architecture, the system should use some tools to analyze and challenge the user's stance. This includes a RAG system to pull verified data from trusted sources, a fact-checker tool to verify the user's claims and a fallacy detector to identify logical fallacies (like strawman arguments or personal attacks). The model will be fine-tuned on debate transcripts to create responses using structured debate mechanics (claim, evidence, impact) and will use RLHF/DPO to punish hallucinated facts and to make sure the AI remains aggressively logical without becoming toxic.

The result is an app where users can practice debating and get a summary of their strengths and weaknesses at the end of the debate.

Motivation

The ability to create a sound argument and defend a point of view is an essential, foundational skill for nearly every aspect of life. But it is a skill that is hard to practice outside of real-world situations. Whether an individual is trying to negotiate for something they want, explain the motivation behind a specific request (such as pitching this very project proposal) or argue why a critical business decision should be taken, success depends entirely on the ability to persuade. True leadership requires the capacity to create and express complex opinions clearly so that others can understand the vision and feel confident following it.

Despite the importance of this skill, people rarely have access to a knowledgeable, objective sparring partner willing to rigorously challenge their ideas before an important conversation. The Devil's Advocate should help the process of learning *how* to argue. By forcing users to defend their stances against a relentless, fact-driven opponent, the application helps users find blind spots in their logic, refine their rhetoric and become more compelling and confident communicators.

Most people currently just use general AI like ChatGPT-5 or Claude 3.5 to roleplay arguments, though these aren't specifically built for debating. Model Diplomat is one of the top specialized tools right now, using an agentic approach to help users practice against AI that's designed to find flaws in their logic. Other platforms like SchoolAI have created designed spaces where students can go back and forth with the AI to sharpen their critical thinking skills before a real-world talk. Even though IBM's famous Project Debater has shifted its focus, it really set the stage for how these newer tools use "Key Point Analysis" to break down complex arguments.